

6.4 U-D FILTERING

U-D filtering was introduced in [Bie76, Bie77a] as another way to increase the numerical precision of the Kalman filter. It is sometimes considered as a type of square root filtering, and sometimes it is considered distinct from square root filtering (depending on the author). It increases the computational cost of the filter but not so severely as the square root filter of the previous section.

The idea of U-D filtering is to factor the $n \times n$ matrix P as UDU^T , where U is an $n \times n$ upper triangular matrix with ones along the diagonal, and D is an $n \times n$ diagonal matrix. This can always be accomplished for a symmetric positive definite matrix P [Gol89, Chapter 4], so it can always be implemented on a Kalman filter. A U-D factorization routine can be implemented without too much difficulty. For example, suppose that we want to compute the U-D factorization of a 3×3 matrix. We can then write

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{12} & p_{22} & p_{23} \\ p_{13} & p_{23} & p_{33} \end{bmatrix} = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ u_{12} & 1 & 0 \\ u_{13} & u_{23} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} d_{11} + d_{22}u_{12}^2 + d_{33}u_{13}^2 & d_{22}u_{12} + d_{33}u_{13}u_{23} & d_{33}u_{13} \\ d_{22}u_{12} + d_{33}u_{13}u_{23} & d_{22} + d_{33}u_{23}^2 & d_{33}u_{23} \\ d_{33}u_{13} & d_{33}u_{23} & d_{33} \end{bmatrix} \quad (6.113)$$

We need to solve for the u_{ij} and d_{ii} elements. We can begin at the lower-right element of the matrix equality to see that $d_{33} = p_{33}$. Next we can look at the other elements in the third column to see that

$$\begin{aligned} u_{13} &= p_{13}/d_{33} \\ u_{23} &= p_{23}/d_{33} \end{aligned} \quad (6.114)$$

Now look at the (2, 2) and (1, 2) elements of the equality to see that

$$\begin{aligned} d_{22} &= p_{22} - d_{33}u_{23}^2 \\ u_{12} &= (p_{12} - d_{33}u_{13}u_{23})/d_{22} \end{aligned} \quad (6.115)$$

Finally look at the (1, 1) element of the equality to see that

$$d_{11} = p_{11} - d_{22}u_{12}^2 - d_{33}u_{13}^2 \quad (6.116)$$

This gives us the U-D factorization for a 3×3 symmetric matrix, and provides the outline for a general U-D factorization algorithm.

6.4.1 U-D filtering: The measurement-update equation

Recall from Equation (5.19) the measurement update equation for the covariance of the Kalman filter:

$$P^+ = P^- - P^-H^T(HP^-H^T + R)^{-1}HP^- \quad (6.117)$$

We have omitted the time subscripts for ease of notation. Now suppose that we process the measurements sequentially as discussed in Section 6.1. This gives the equation

$$P_i = P_{i-1} - P_{i-1}H_i^T(H_iP_{i-1}H_i^T + R_i)^{-1}H_iP_{i-1} \quad (6.118)$$

where H_i is the i th row of H , R_i is the i th diagonal entry of R , and P_i is the estimation covariance after i measurements have been processed. Now define the scalar $\alpha_i \equiv H_i P_{i-1} H_i^T + R_i$. Suppose that $P_{i-1} = U_{i-1} D_{i-1} U_{i-1}^T$, and $P_i = U_i D_i U_i^T$. With these factorizations we can write the measurement update of Equation (6.118) as

$$\begin{aligned} U_i D_i U_i^T &= U_{i-1} D_{i-1} U_{i-1}^T - \frac{1}{\alpha_i} U_{i-1} D_{i-1} U_{i-1}^T H_i^T H_i U_{i-1} D_{i-1} U_{i-1}^T \\ &= U_{i-1} \left[D_{i-1} - \frac{1}{\alpha_i} (D_{i-1} U_{i-1}^T H_i^T) (D_{i-1} U_{i-1}^T H_i^T)^T \right] U_{i-1}^T \end{aligned} \quad (6.119)$$

The term in brackets in the above equation is symmetric positive definite so it has a U-D factorization that can be written as

$$\bar{U} \bar{D} \bar{U}^T = \left[D_{i-1} - \frac{1}{\alpha_i} (D_{i-1} U_{i-1}^T H_i^T) (D_{i-1} U_{i-1}^T H_i^T)^T \right] \quad (6.120)$$

Combining this with Equation (6.119) gives

$$\begin{aligned} U_i D_i U_i^T &= U_{i-1} \bar{U} \bar{D} \bar{U}^T U_{i-1}^T \\ &= (U_{i-1} \bar{U}) \bar{D} (U_{i-1} \bar{U})^T \end{aligned} \quad (6.121)$$

Note that $U_{i-1} \bar{U}$ is upper triangular with diagonal elements equal to 1, and $\bar{D}$ is diagonal. Therefore the above equation means that $U_i = U_{i-1} \bar{U}$, and $D_i = \bar{D}$:

$$\begin{aligned} U_i &= U_{i-1} \bar{U} \\ D_i &= \bar{D} \end{aligned} \quad (6.122)$$

This gives us a way of performing the measurement update of P in terms of its U-D factors. The algorithm can be summarized as follows.

The U-D measurement update

1. We start with the *a priori* estimation covariance P^- at time k . Define $P_0 = P^-$.
2. For $i = 1, \dots, r$ (where r is the number of measurements), perform the following:
 - (a) Define H_i as the i th row of H , R_i as the i th diagonal entry of R , and $\alpha_i = H_i P_{i-1} H_i^T + R_i$.
 - (b) Perform a U-D factorization of P_{i-1} to obtain U_{i-1} and D_{i-1} , and then form the matrix on the right side of Equation (6.120).
 - (c) Find the U-D factorization of the matrix on the right side of Equation (6.120) and call the factors $\bar{U}$ and $\bar{D}$.
 - (d) Compute U_i and D_i from Equation (6.122).
3. The *a posteriori* estimation covariance is given as $P^+ = U_r D_r U_r^T$.

Since the U-D measurement-update equation relies on sequential filtering, the conditions discussed at the end of Section 6.1 apply to U-D filtering. That is, it probably does not make sense to implement U-D filtering unless one of the following two conditions is true.

1. The measurement noise covariance R_k is diagonal
2. The measurement noise covariance R is a constant.

6.4.2 U-D filtering: The time-update equation

Recall from Equation (5.19) the time-update equation for the covariance of the Kalman filter:

$$P^- = FP^+F^T + Q \quad (6.123)$$

We have omitted the time subscripts for ease of notation. If the Kalman filter is being used to estimate the state of an n -state system, then the P matrices will be $n \times n$ matrices. Suppose that P^+ is factored as $U^+D^+U^{+T}$ (from the measurement update equation discussed previously). We need to find the U-D factors of P^- such that $P^- = U^-D^-U^{-T} = FP^+F^T + Q$. Note that U^{-T} in this notation is *not* the transpose of the inverse of U ; it is rather the transpose of U^- . The time update of Equation (6.123) can be written as

$$\begin{aligned} P^- &= FP^+F^T + Q \\ &= [FU^+ \quad I] \begin{bmatrix} D^+ & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} U^{+T}F^T \\ I \end{bmatrix} \\ &= W\hat{D}W^T \end{aligned} \quad (6.124)$$

where W and $\hat{D}$ are defined by the above equation. Note that W is an $n \times 2n$ matrix, and $\hat{D}$ is a $2n \times 2n$ matrix. From the above equation we see that the U-D factors of P^- need to satisfy

$$U^-D^-U^{-T} = W\hat{D}W^T \quad (6.125)$$

The transpose of W can be written as

$$W^T = [w_1^T \quad \cdots \quad w_n^T] \quad (6.126)$$

That is, w_i (a $2n$ -element row vector) is the i th row of W . Now we find n vectors v_i such that

$$v_k\hat{D}v_j^T = 0 \quad k \neq j \quad (6.127)$$

The v_i vectors ($2n$ -element row vectors) can be found with the following Gram-Schmidt orthogonalization procedure [Hor85, Gol89, Moo00]:

$$\begin{aligned} v_n &= w_n \\ v_k &= w_k - \sum_{j=k+1}^n \frac{w_k\hat{D}v_j^T}{v_j\hat{D}v_j^T} v_j \quad k = n-1, \dots, 1 \end{aligned} \quad (6.128)$$

If we define $u(k, j)$ as

$$u(k, j) = \frac{w_k\hat{D}v_j^T}{v_j\hat{D}v_j^T} \quad j, k = 1, \dots, n \quad (6.129)$$

then from Equation (6.128) we see that w_k can be expressed as

$$w_k = v_k + \sum_{j=k+1}^n u(k, j)v_j \quad k = 1, \dots, n \quad (6.130)$$

or equivalently

$$w_k^T = v_k^T + \sum_{j=k+1}^n u(k, j)v_j^T \quad k = 1, \dots, n \quad (6.131)$$

These n equations can be written as

$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} 1 & u(1, 2) & \cdots & u(1, n) \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & u(n-1, n) \\ 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$W = U^-V \quad (6.132)$$

The $n \times 2n$ matrix W , the $n \times n$ matrix U^- , and the $n \times 2n$ matrix V are defined by the above equation. Note that U^- is a unit upper triangular matrix. The matrix product $W\hat{D}W^T$ can then be written as

$$\begin{aligned} W\hat{D}W^T &= (U^-V)\hat{D}(U^-V)^T \\ &= U^-(V\hat{D}V^T)U^{-T} \\ &= U^-D^-U^{-T} \end{aligned} \quad (6.133)$$

where the D^- matrix is defined by the above equation. From Equation (6.127), we see that the v_i vectors are orthogonal with respect to the $\hat{D}$ inner product. We therefore know that

$$\begin{aligned} D^- &= V\hat{D}V^T = \text{diag}(d_1, \dots, d_n) \\ d_k &= v_k\hat{D}v_k^T \end{aligned} \quad (6.134)$$

That is, D^- is a diagonal matrix. From Equations (6.124), (6.125), and (6.133) we see that U^- and D^- satisfy the conditions of being the U-D factors of P^- . This gives us a way to perform the Kalman filter time-update equation in U-D factorization form. The algorithm can be summarized as follows.

The U-D time update

1. Begin with $P^+ = U^+D^+U^{+T}$ (from the measurement update equation).
2. Define the following matrices.

$$\begin{aligned} W &= \begin{bmatrix} FU^+ & I \end{bmatrix} \\ \hat{D} &= \begin{bmatrix} D^+ & 0 \\ 0 & Q \end{bmatrix} \end{aligned} \quad (6.135)$$

3. Use the rows of W along with the Gram-Schmidt orthogonalization procedure to generate v_i vectors that are orthogonal with respect to the $\hat{D}$ inner product. The algorithm for generating the v_i vectors is given in Equation (6.128).